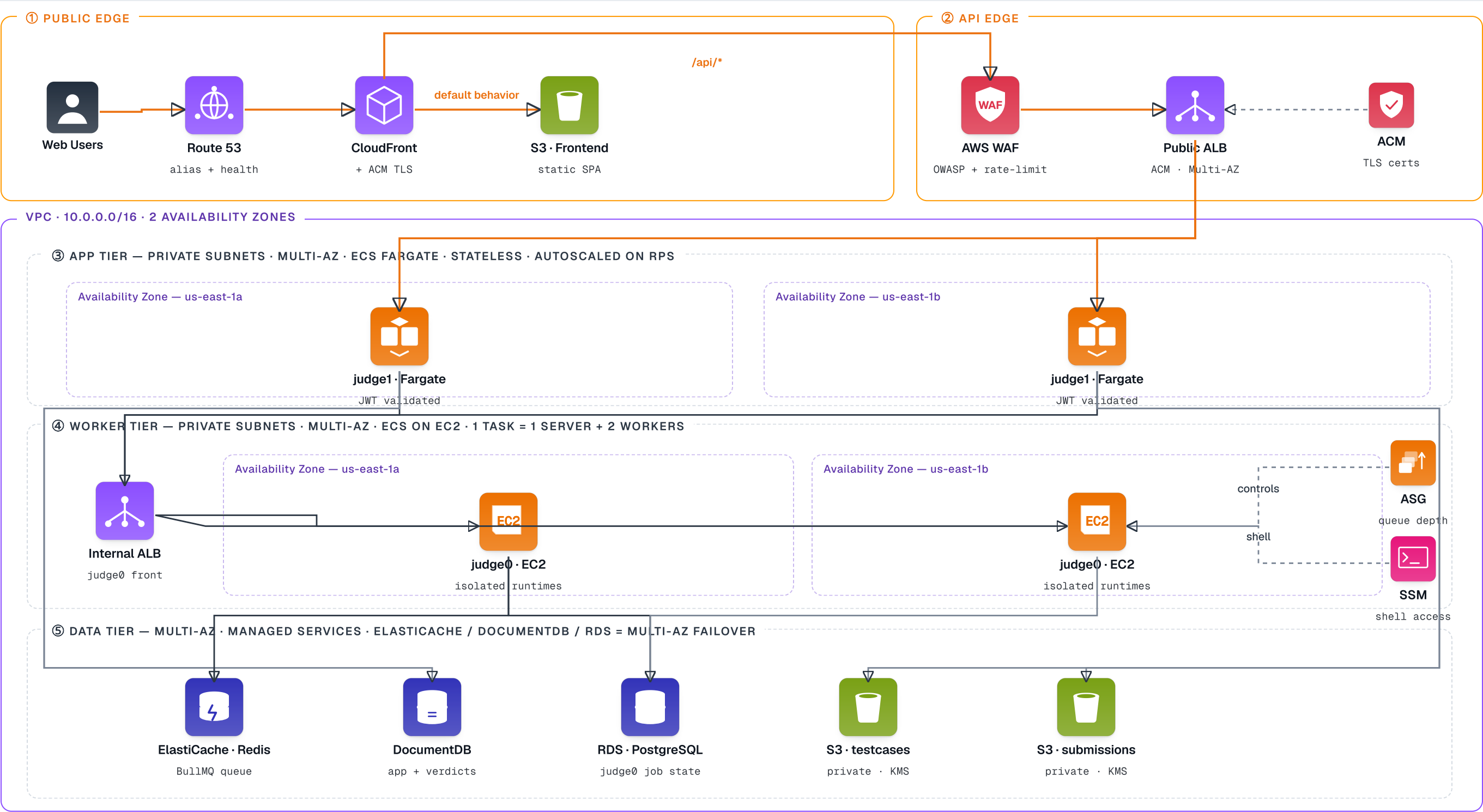


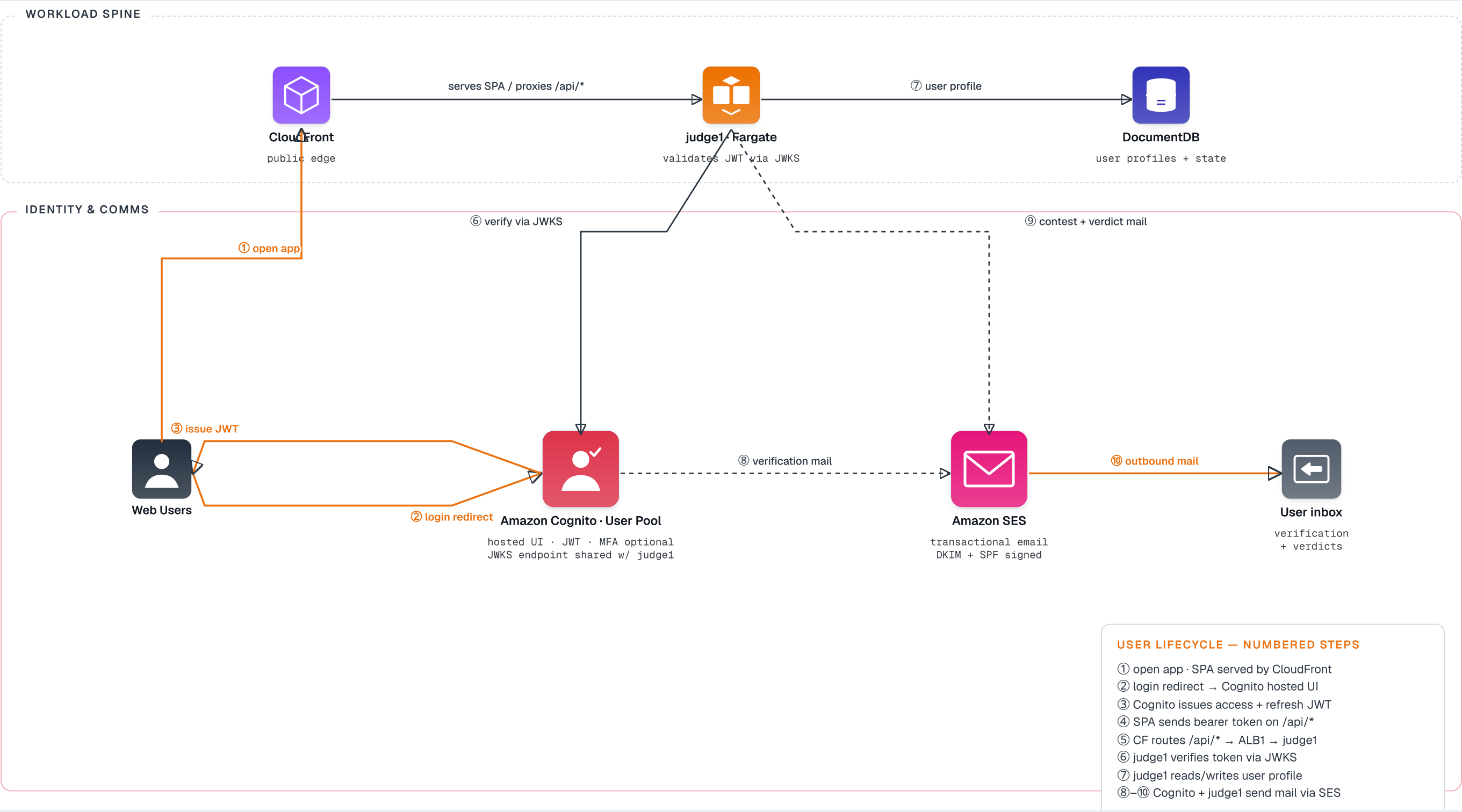
Request Lifecycle

HTTPS lands at Route 53 → CloudFront. The default behavior serves the SPA from S3; /api/* routes through WAF and the public ALB into the multi-AZ Fargate app tier, which fans out to the EC2 worker pool and the managed data stores.



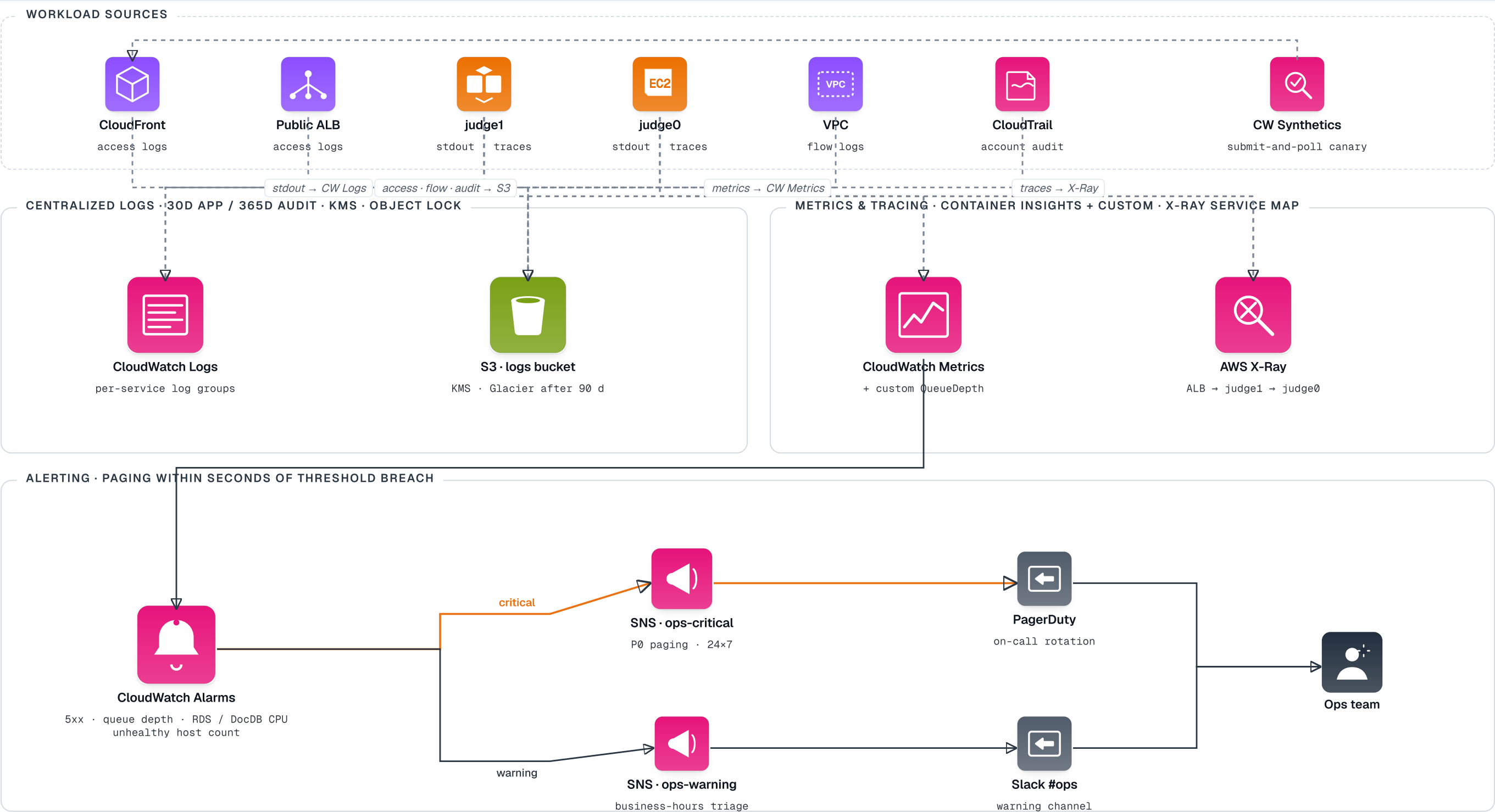
Identity & User Lifecycle

Cognito User Pool owns user records and issues JWTs. Browsers authenticate against the hosted UI; judge1 validates each bearer token via JWKS. Cognito and judge1 both push transactional email through SES.



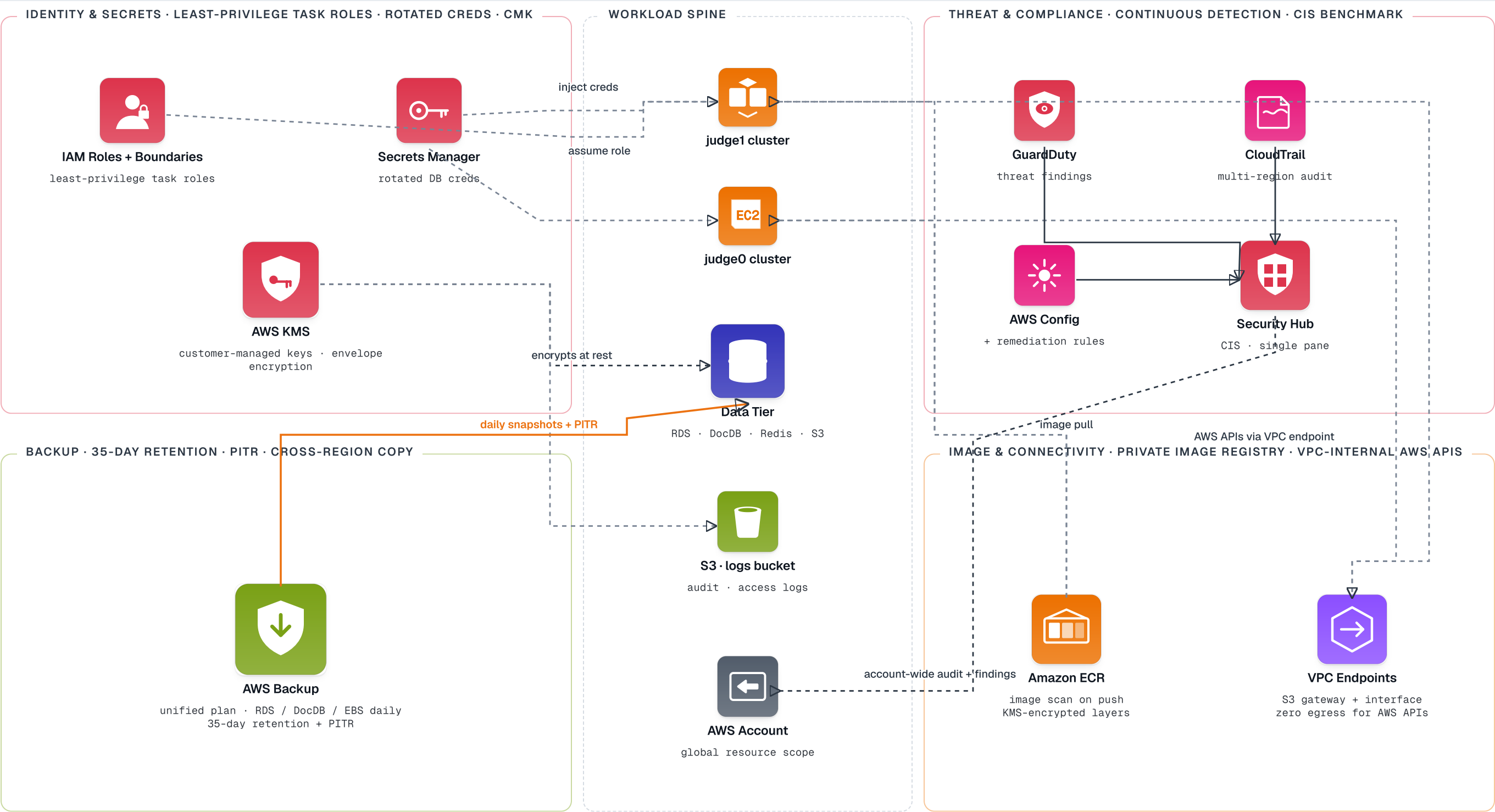
Monitoring, Logging & Alerting

Every workload component ships structured stdout to CloudWatch Logs and traces to X-Ray. Access logs flow to S3. Metrics drive CloudWatch Alarms which split critical vs. warning across two SNS topics, paging Ops in seconds.



Security, Backup & Supporting Services

Every component is identity-scoped and key-encrypted. Account-wide threat detection feeds a single Security-Hub pane. AWS Backup snapshots the data tier daily. ECR + VPC endpoints keep image pulls and AWS-API traffic on the private backbone.



Single-Submission Flow

From the moment a contestant hits "Submit" until the final verdict streams back over WebSocket. judge1 owns orchestration: it persists the submission, queues the job, fetches testcases from S3, dispatches batches to judge0, polls for results, persists the verdict, and pushes it back over a WebSocket.

